Exact Trig Values

Exam code: 0607

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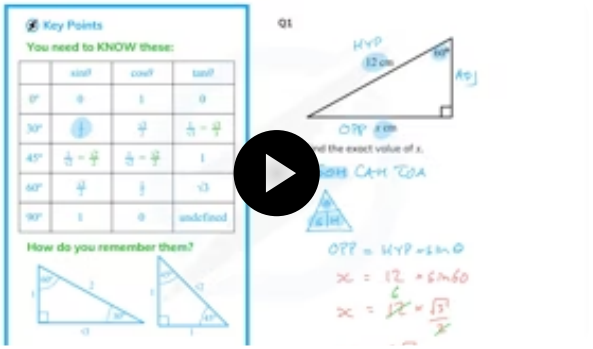
Updated on 18 February 2026



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Help me understand

Exact trig values

What are exact values in trigonometry?

- For **certain angles** the values of sin θ, cos θ and tan θ can be written **exactly**
  - This means using fractions and surds
- You are **expected to know** the exact values of sin, cos and tan for
  - 0°, 30°, 45°, 60°, 90°, 180° and their multiples

θ	0°	30°	45°	60°	90°
sin θ	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos θ	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan θ	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

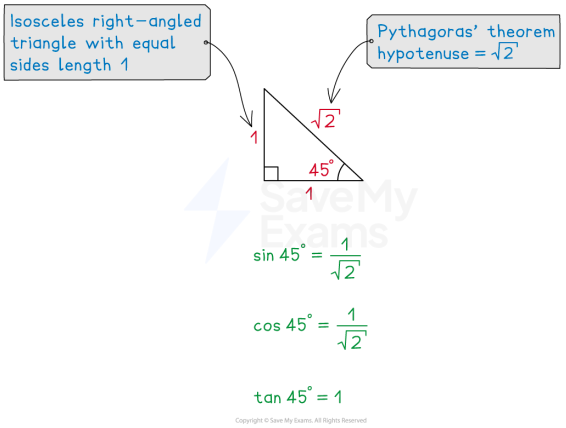
- Look at **patterns** in the table
  - Note the values of sin θ from 0° to 90° match cos θ in reverse, from 90° to 0°

Examiner Tips and Tricks

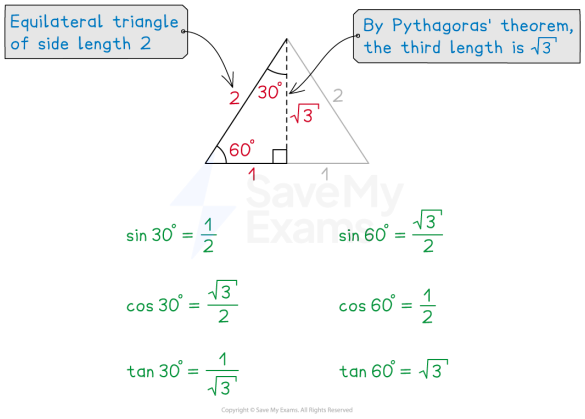
Some people remember sin θ using the trick $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$ which simplifies to $0, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$.

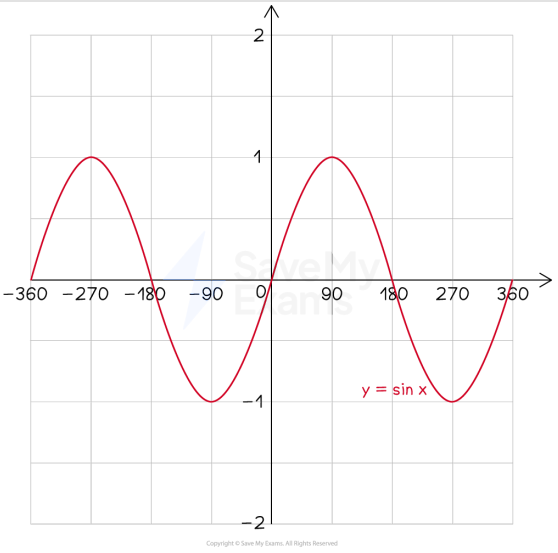
- Two special **right-angled triangles** below can help you to find the exact values for 30°, 45° and 60°
  - Remember that by rationalising the denominator, $\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

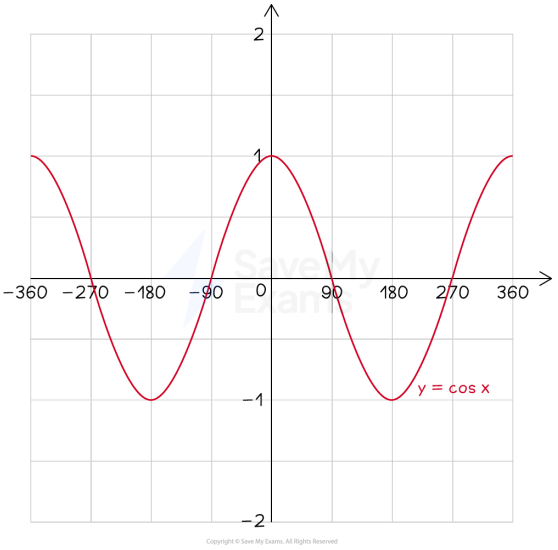
Triangle for 45°

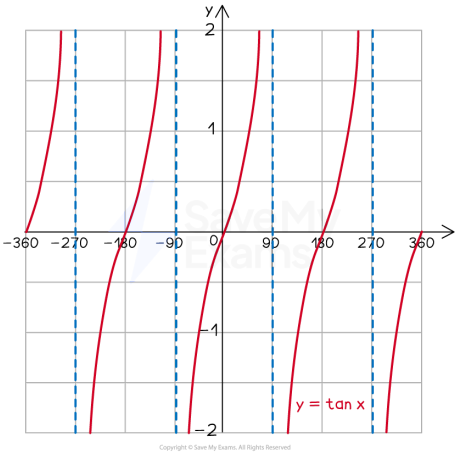


Triangle for 30° and 60°









How do I use exact trig values?

- You may come across trig questions in a **non-calculator** question

- E.g. Solve the equation $\cos 45 = \frac{x}{12}$
  - Replace $\cos 45$ with $\frac{\sqrt{2}}{2}$ to give $\frac{\sqrt{2}}{2} = \frac{x}{12}$
  - Then you can solve for x

- On trig graphs, you may be expected to find a coordinate
  - E.g. The coordinates $(30, k)$ lie on the graph $y = \tan x$, find k
    - k will be equal to $\tan 30$
    - The exact value of $\tan 30$ is $\frac{\sqrt{3}}{3}$
    - Therefore $k = \frac{\sqrt{3}}{3}$

Examiner Tips and Tricks

- Writing these out (or sketching the triangles/graphs) on your paper at the beginning of the exam means that you can use them as many times as you need to during the exam!

Worked Example

Find the value of x in the diagram below.

Give your answer as an exact value.

Answer:

Triangle *ABC* is a right-angled triangle, so use SOHCAHTOA

We know the hypotenuse (*AC*) and we want to calculate the opposite (*BC*), so use $\sin \theta = \frac{\text{O}}{\text{H}}$

$$\sin 60 = \frac{x}{28}$$

Remember that $\sin 60 = \frac{\sqrt{3}}{2}$

So,

$$28 \times \frac{\sqrt{3}}{2} = x$$

$$14 \times \sqrt{3} = x$$

Leave in exact (surd) form

$$x = 14\sqrt{3} \text{ cm}$$

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